Failure of stable CSM positivity at the empty partition

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Task 1. Resolve the shifted-binomial positivity and log-concavity assertion for stable CSM–Schur coefficients in [1, Conjecture 6.7]. For a partition λ whose parts are at least 2, put $c = |\lambda| - \ell(\lambda)$ and write

$$c^{\mathrm{SM}}(Y_\lambda(d))_f = \sum_{i=0}^{\lfloor f/2 \rfloor} C_{\lambda,f,i}(d) s_{f-i,i}(a,b), \qquad f \geq c.$$

Here $Y_\lambda(d)$ is the *open* coincident-root stratum. With $\widetilde{C}_{\lambda,f,i} = (-1)^{f-c} C_{\lambda,f,i}$, the assertion requires all coefficients γ_j in

$$\widetilde{C}_{\lambda,f,i}\big(y + |\lambda| + 2(f-c)\big) = \sum_{j \geq 0} \gamma_j \binom{y}{j}$$

to be nonnegative, with the resulting sequence log-concave. As formulated, this includes the empty partition: it has no forbidden parts, and the source explicitly discusses it immediately before the conjecture. This case already contradicts the positivity requirement.

Proposition 1. *For the empty partition, $f = 1$, and $i = 0$, the stable coefficient is*

$$C_{\varnothing,1,0}(d) = \frac{d(3-d)}{2}.$$

Consequently the conjecture's shifted signed polynomial is

$$\widetilde{C}_{\varnothing,1,0}(y+2) = \frac{(y+2)(y-1)}{2} = -\binom{y}{0} + \binom{y}{1} + \binom{y}{2}.$$

In particular, $\gamma_0 = -1 < 0$.

Proof. Let $V_d = \mathrm{Pol}^d(\mathbb{C}^2)$, using the source's positive-weight convention. Its weights are $ja + (d-j)b$ for $0 \leq j \leq d$, and therefore

$$c_1(V_d) = \sum_{j=0}^{d} (ja + (d-j)b) = \frac{d(d+1)}{2}(a+b).$$

For every $d \geq 2$, the open stratum $Y_\varnothing(d)$ is the complement of the binary discriminant hypersurface D_d. The discriminant has character $(\det)^{d(d-1)}$, so

$$[D_d] = d(d-1)(a+b).$$

For completeness, write a split form as $\prod_{r=1}^{d} \ell_r$. Its discriminant is, up to a nonzero constant, $\prod_{r<s} \det(\ell_r, \ell_s)^2$. A linear change of variables multiplies each determinant by det, which proves

the stated character. The discriminant hypersurface is irreducible, being the closure of the forms $\ell^2 g$ with one double root. It is reduced: along $(X^2 - tY^2)g(X, Y)$ with the roots of g distinct and away from $X = 0$, its discriminant is t times a function nonzero at $t = 0$.

The normalization and additivity of equivariant CSM classes give

$$c^{\mathrm{SM}}(V_d) = c(V_d), \qquad c^{\mathrm{SM}}(Y_\varnothing(d)) = c^{\mathrm{SM}}(V_d) - c^{\mathrm{SM}}(D_d).$$

The codimension-one part of $c^{\mathrm{SM}}(D_d)$ is its fundamental class. Thus

$$c^{\mathrm{SM}}(Y_\varnothing(d))_1 = c_1(V_d) - [D_d] = \frac{d(3-d)}{2}(a+b).$$

Since $s_{1,0}(a, b) = a + b$, this identifies the Schur coefficient for every $d \geq 2$, and hence identifies its stable polynomial. Now $c = 0$, the sign is -1, and the prescribed shift is $y = d-2$. Substitution gives the displayed binomial expansion. Alternatively, evaluation at $y = 0$ already forces its constant binomial coefficient to be -1. □

This is a counterexample to the universal formulation, not a resolution of the conjecture after imposing $\lambda \neq \varnothing$. The distinction between the open stratum and its closure is essential: $\overline{Y_\varnothing(d)} = V_d$, whereas $Y_\varnothing(d)$ consists of squarefree forms. The source also distinguishes these objects after Conjecture 6.7. No assertion about classes of closures is used here.

Task 2. Determine whether both conclusions of Conjecture 6.7 hold for every nonempty partition λ with parts at least 2.

Task 3. For $\lambda = \varnothing$, determine the stable Schur coefficient polynomials in all homogeneous degrees and the least integer shifts that make their signed binomial coefficients nonnegative.

Task 4. For those corrected empty-partition shifts, determine which coefficient sequences are log-concave and classify any exceptions.

Task 5. Formulate and test a separate binomial positivity and log-concavity statement for CSM classes of coincident-root stratum closures, keeping their normalization distinct from that of open strata.

References

[1] G. Bérczi and L. M. Fehér, *Positivity in classical enumerative geometry: a case study in synchronized AI-assisted mathematics*, arXiv:2605.25271v1 (2026), Section 6.2, pp. 26–27. `https://arxiv.org/abs/2605.25271`.

[2] A. Juhász, *Equivariant classes of coincident root strata and related invariants of varieties of tangent lines*, doctoral dissertation, Central European University (2023), Chapter 3, in particular Theorem 3.6.1. `https://hdl.handle.net/20.500.14018/29523`.